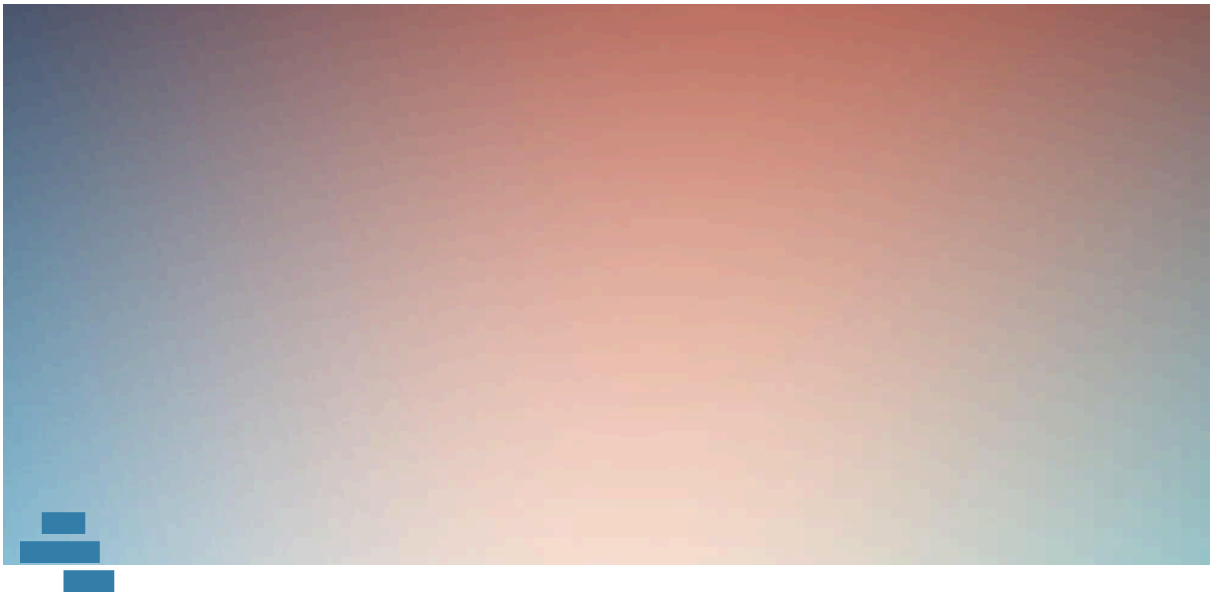
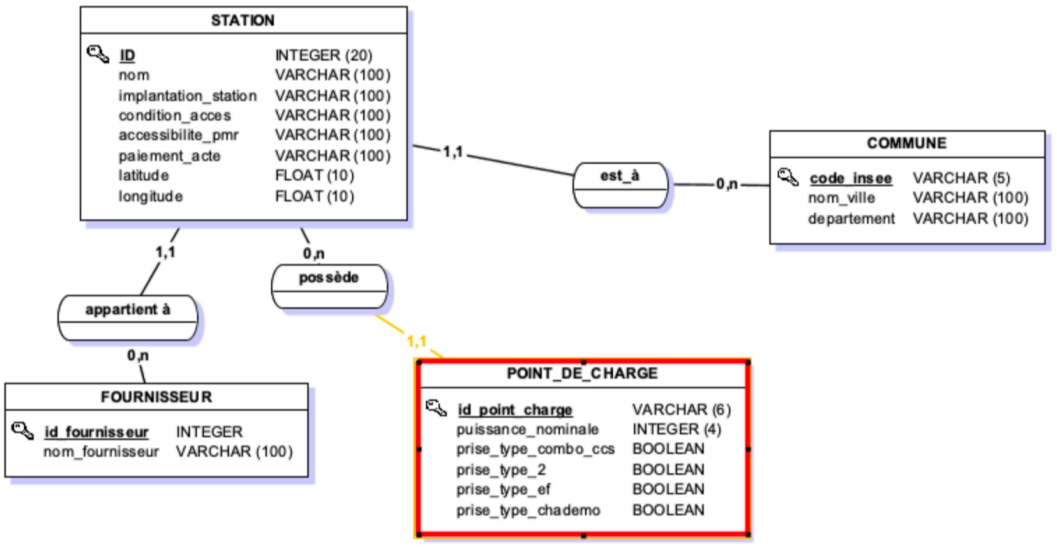


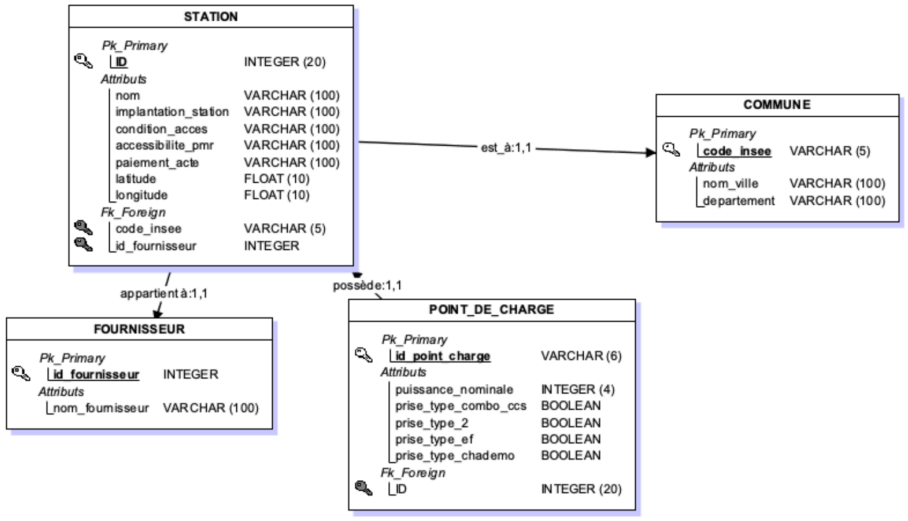
Modèle conceptuel de Donnée

Catégorie	Analyse
Date	June 15, 2026
Personne	Elouen NAEL
Sélectionner	Après-midi
État	Terminé

MCD:



MLD:




```sql
DROP DATABASE IF EXISTS irve;
CREATE DATABASE irve CHARACTER SET utf8mb4 COLLATE utf8mb4_
unicode_ci;
USE irve;

-- ------------------------------
```

```sql
-- Table: COMMUNE
-- ----------------------------
CREATE TABLE COMMUNE (
  code_insee   VARCHAR(5)   NOT NULL,
  nom_ville    VARCHAR(100) NOT NULL,
  departement  VARCHAR(100) NOT NULL,
  CONSTRAINT COMMUNE_PK PRIMARY KEY (code_insee)
) ENGINE=InnoDB;

-- ----------------------------
-- Table: FOURNISSEUR
-- ----------------------------
CREATE TABLE FOURNISSEUR (
  id_fournisseur INT NOT NULL AUTO_INCREMENT,
  nom_fournisseur VARCHAR(100) NOT NULL,
  CONSTRAINT FOURNISSEUR_PK PRIMARY KEY (id_fournisseur)
) ENGINE=InnoDB;

-- ----------------------------
-- Table: STATION
-- ----------------------------
CREATE TABLE STATION (
  ID INT NOT NULL AUTO_INCREMENT,
  nom                  VARCHAR(100) NOT NULL,
  implantation_station  VARCHAR(100) NOT NULL,
  condition_acces      VARCHAR(100) NOT NULL,
  accessibilite_pmr    VARCHAR(100) NOT NULL,
  paiement_acte        VARCHAR(100) NOT NULL,
  latitude             FLOAT        NOT NULL,
  longitude            FLOAT        NOT NULL,
  code_insee           VARCHAR(5)   NOT NULL,
  id_fournisseur       INT          NOT NULL,

  CONSTRAINT STATION_PK PRIMARY KEY (ID),
  CONSTRAINT STATION_code_insee_FK
    FOREIGN KEY (code_insee) REFERENCES COMMUNE(code_inse
e),
  CONSTRAINT STATION_id_fournisseur_FK
```

```
    FOREIGN KEY (id_fournisseur) REFERENCES FOURNISSEUR(id_
fournisseur)
) ENGINE=InnoDB;

-- ----------------------------
-- Table: POINT_DE_CHARGE
-- ----------------------------
CREATE TABLE POINT_DE_CHARGE (
  id_point_charge      VARCHAR(6)  NOT NULL,
  puissance_nominale   INT         NOT NULL,
  prise_type_combo_ccs TINYINT(1)  NOT NULL,
  prise_type_2         TINYINT(1)  NOT NULL,
  prise_type_ef        TINYINT(1)  NOT NULL,
  prise_type_chademo   TINYINT(1)  NOT NULL,

  ID_station           INT         NOT NULL,

  CONSTRAINT POINT_DE_CHARGE_PK PRIMARY KEY (id_point_charg
e),
  CONSTRAINT POINT_DE_CHARGE_station_FK
    FOREIGN KEY (ID_station) REFERENCES STATION(ID)
) ENGINE=InnoDB;

-- ===== INDEX (utiles pour les jointures) =====
-- (PK déjà indexées automatiquement)
CREATE INDEX STATION_code_insee_IDX ON STATION (code_inse
e);
CREATE INDEX STATION_id_fournisseur_IDX ON STATION (id_four
nisseur);
CREATE INDEX PDC_ID_station_IDX ON POINT_DE_CHARGE (ID_stat
ion);
```

```
USE irve;

INSERT INTO COMMUNE (code_insee, nom_ville, departement) VA
LUES
('29232', 'Brest', 'Finistère'),
('75056', 'Paris', 'Paris');
```

```sql
INSERT INTO FOURNISSEUR (id_fournisseur, nom_fournisseur) V
ALUES
(1, 'Izivia'),
(2, 'TotalEnergies');

INSERT INTO STATION (
  nom, implantation_station, condition_acces, accessibilite
_pmr, paiement_acte,
  latitude, longitude, code_insee, id_fournisseur
) VALUES
('Station Brest Centre', 'Rue de Siam', 'Accès public', 'Ou
i', 'CB',
 48.3904, -4.4861, '29232', 1),
('Station Paris 1', 'Châtelet', 'Accès public', 'Oui', 'C
B',
 48.8589, 2.3470, '75056', 2);

INSERT INTO POINT_DE_CHARGE (
  id_point_charge, puissance_nominale,
  prise_type_combo_ccs, prise_type_2, prise_type_ef, prise_
type_chademo,
  ID_station
) VALUES
('PDC001', 22, 1, 1, 0, 0, 1001),
('PDC002', 50, 1, 0, 0, 1, 1001),
('PDC003', 11, 0, 1, 1, 0, 1002);
```